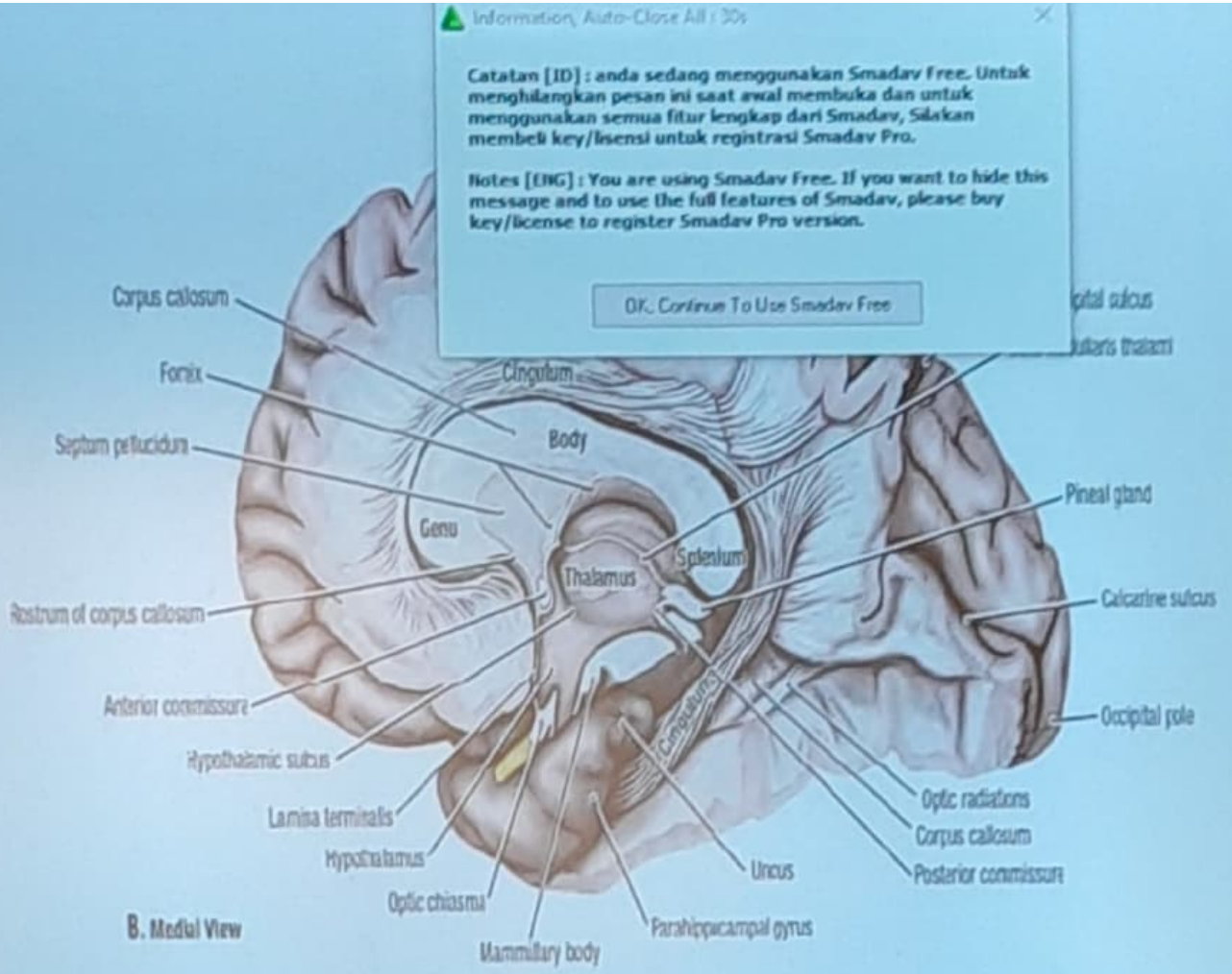
Hypothalamus

Part of diencephalon. Designated as the head ganglion of ANS because of its involvement in controlling many visceral and metabolic activities in the body.

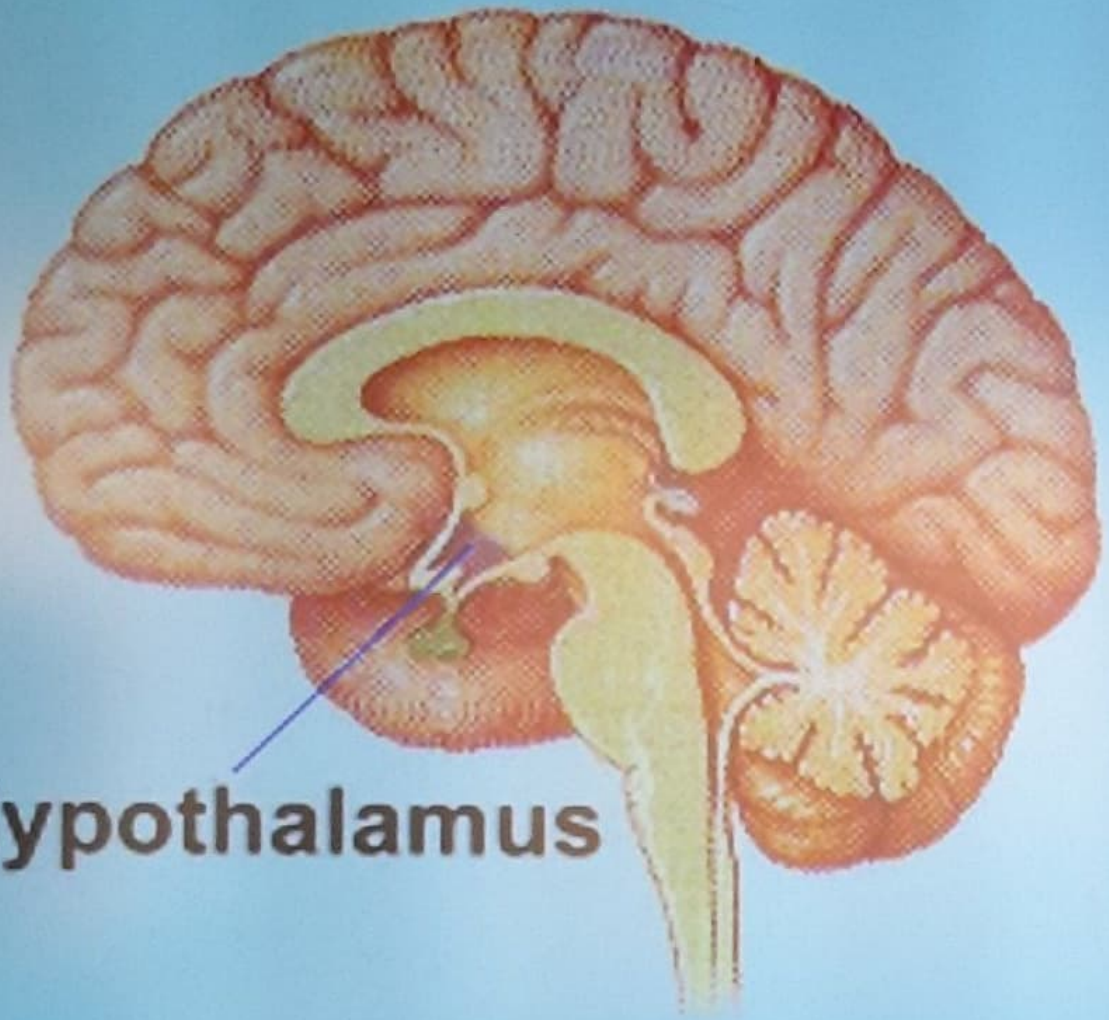
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Anatomically includes; floor of 3rd ventricle, lateral wall of 3rd ventricle below hypothalamic sulcus.

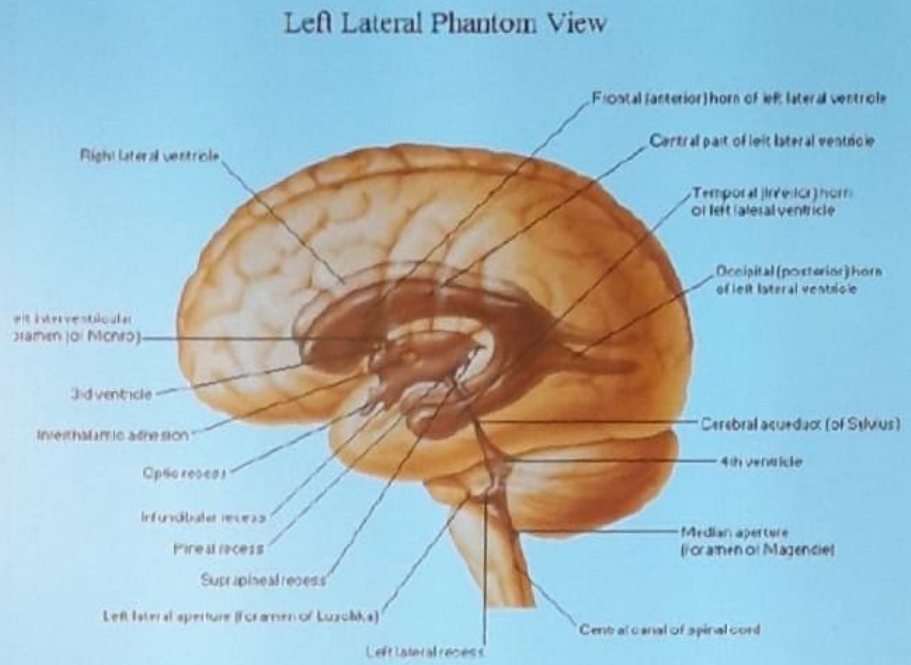
Boundaries: Situated at the base of brain; anteriorly, bounded by lamina terminalis, inferiorly, floor of 3rd ventricle, from optic chiasma, infundibulum, tubercle to posterior perforating substances, superiorly, hypothalamic sulcus.



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Lateral ventricle



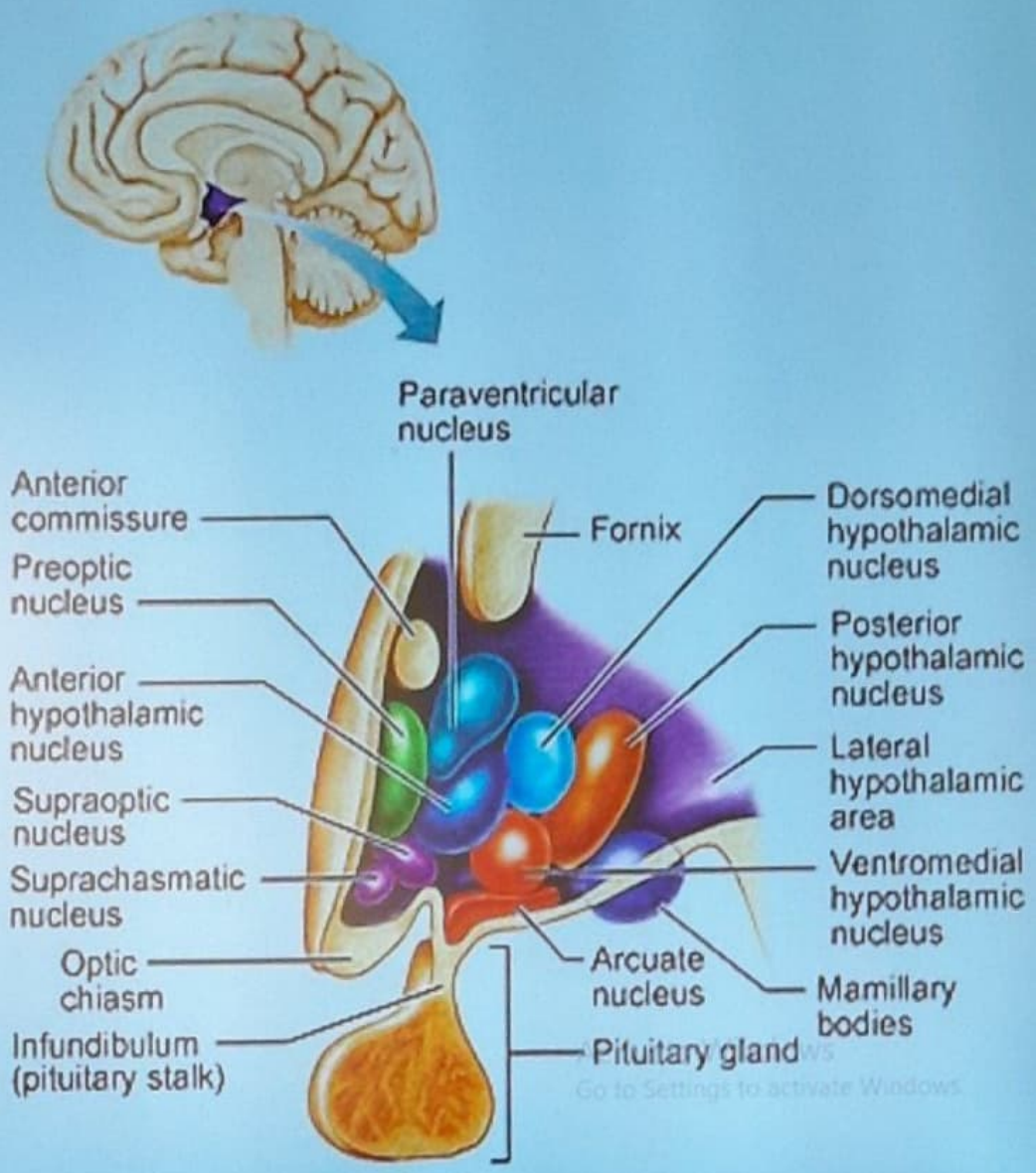
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Parts of Hypothalamus: There are three parts, optic, tuberal and mammary parts.

Optic part: Preoptic is most rostral of the optic group, Paraventricular is above supraoptic nucleus; supraoptic is cranial to suprachiamatic nuclei.

Tuberal part: Ventromedial is dorsal to supraoptic, dorsomedial is in front of thalamus. Tuberal nucleus is lateral to ventromedial nucleus.

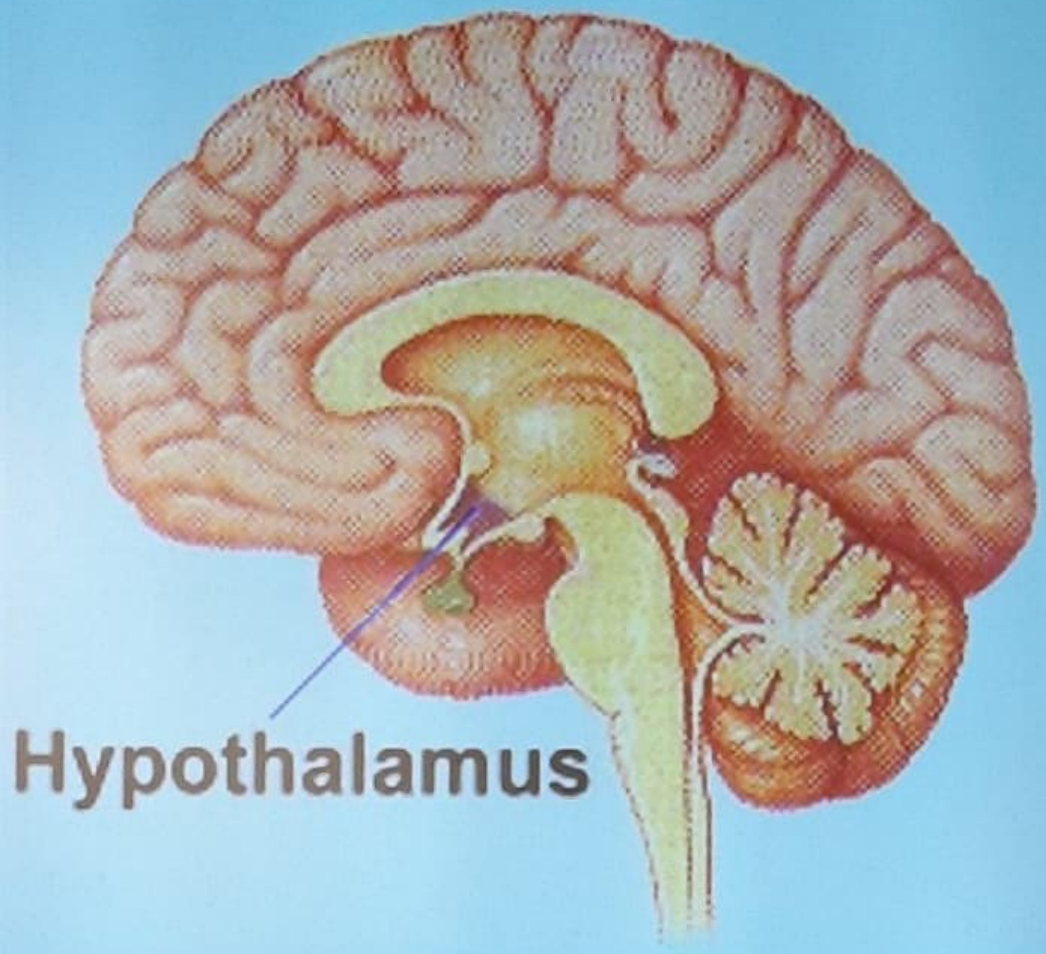
Mammary part: Posterior nuclei is caudal to ventromedial and dorso-medial nucleus, Lateral nucleus is lateral to posterior nucleus.



Autonomic control: Basic autonomic reflexes for respiratory, CVS and GIT does not require suprabulbar control but balanced activity in visceral requires hypothalamic control through descending pathway.

Preoptic control P∑ and posterior hypothalamic control by ∑.

Regulation of body temperature: An important function of preoptic nucleus is that it contains heat sensitive neurons. Rising temperature of arterial blood passes rich vascular preoptic nucleus resulting in arterial dilation while decrease in body temperature initiate heat conservation.



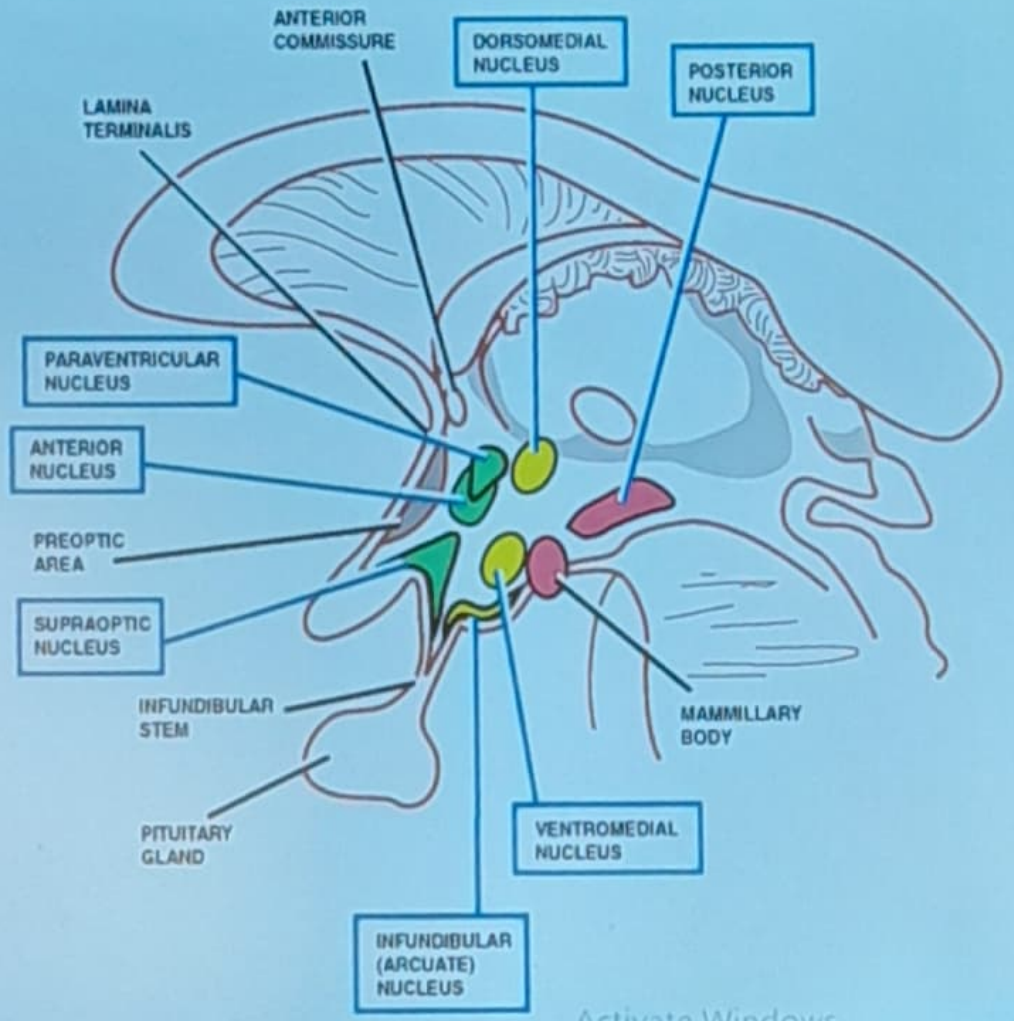
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Endocrine function: Involves secretion of hormone from pituitary body. Supraoptic associated with posterior pituitary while tuber region controls more anteriorly.

Hypothalamus and posterior pituitary:

Hormone ADH or vasopressin is secreted by suprachiasmatic and supraoptic nuclei which regulate body H_2O through tubular absorption.

Oxytocin secreted by paraventricular nucleus leads to uterine contraction and initiation of sucking.



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Hypothalamus and anterior pituitary: The infundibular and ventromedial nuclei of hypothalamus influences the secretion of TSH, TSA, ACTH, FSH, LH, GH, MSH and prolactin from the anterior pituitary.

Regulation of food and water intake: Lateral and medial hypothalamic group of nuclei have opposing actions with regards to eating.

Stimulation of lateral hypothalamic group stimulates eating while stimulation of ventromedial nuclei group will cause the animal to eat less.

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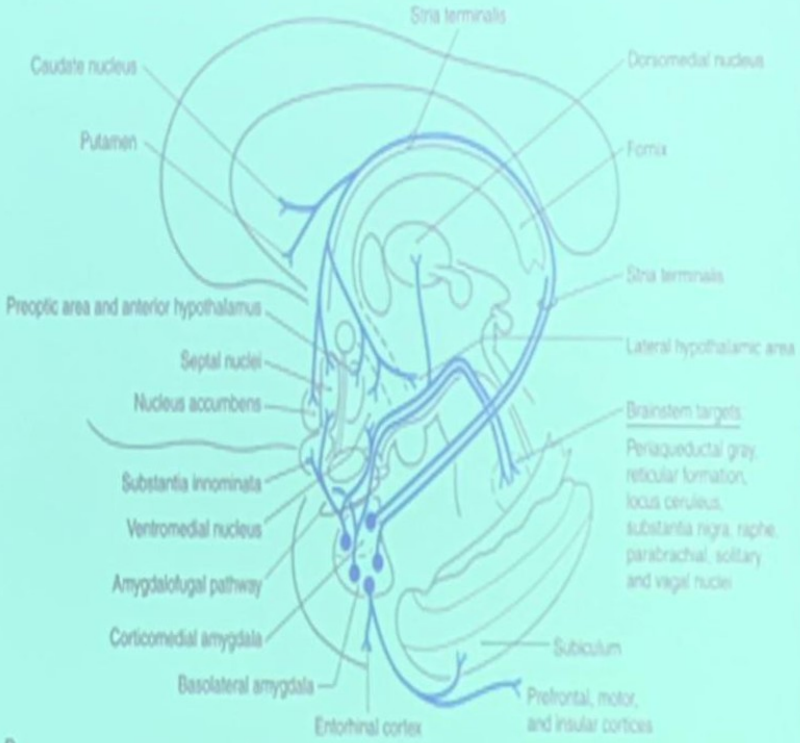
Reproduction and sexual behaviour: It controls reproductive function through gonadotropic hormones of the anterior pituitary. For various emotional aspects of sexual behaviour, close integration with limbic system is involved.

Sexual drive could be stimulated from various areas especially from most anterior and most posterior parts.

Biological clock and circadian rhythms: Body function vary over the 24hrs of the day i.e. fluctuations of body temperature, sleep and wakefulness are influenced by environmental lightening and mediated by suprachiasmatic nuclei.

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Interaction between hypothalamus and limbic system: Amygdala a component of limbic system sends a distress signal to the hypothalamus that usually triggers fight or flight response through the ANS. The hypothalamus acts as a control centre, regulating bodily functions like hormone production, sleep, mood, and autonomic nervous system responses. The amygdala, on the other hand, is crucial for processing emotions, particularly fear and anxiety, and for attaching emotional significance to memories.



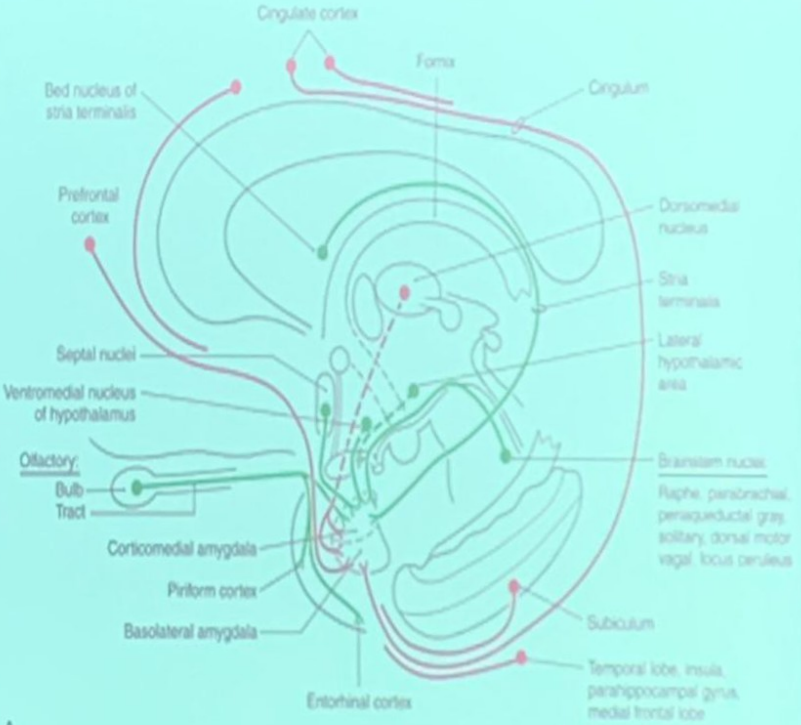
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There pathways of interconnectivity include stria terminalis and the ventral amygdala-fugal pathway which allows bidirectional communication between the amygdala and hypothalamus.

Key pathway and their roles

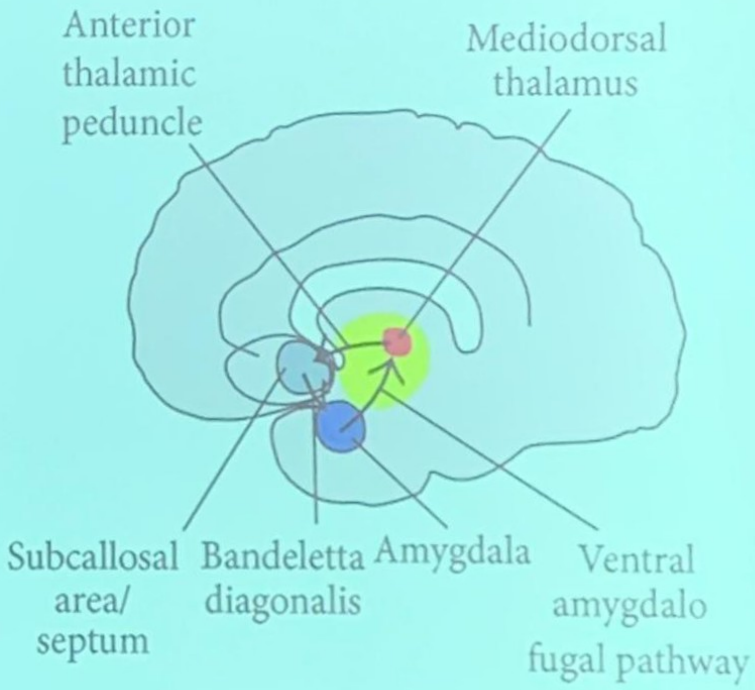
Stria terminalis: The pathway originates from the ventromedial nucleus of amygdala and projects to the ventromedial nucleus of the hypothalamus among other things, it is crucial for various stress responses and can modulate hypothalamic activities.



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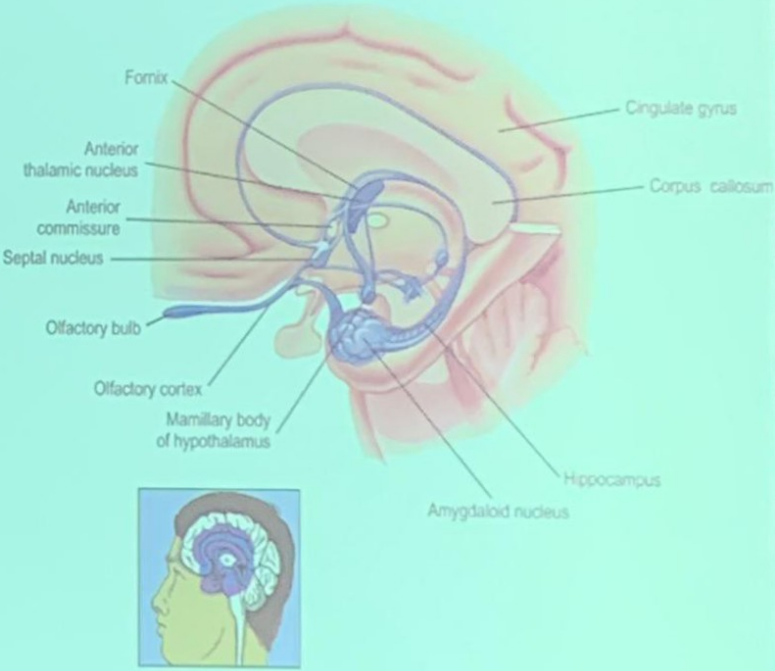
Ventral amygdalofugal pathway:

This pathway with broader reach, that involves septal areas and medial dorsal thalamic nucleus plays a significant role in linking motivation and drives potentially facilitating learned responses and associative learning.

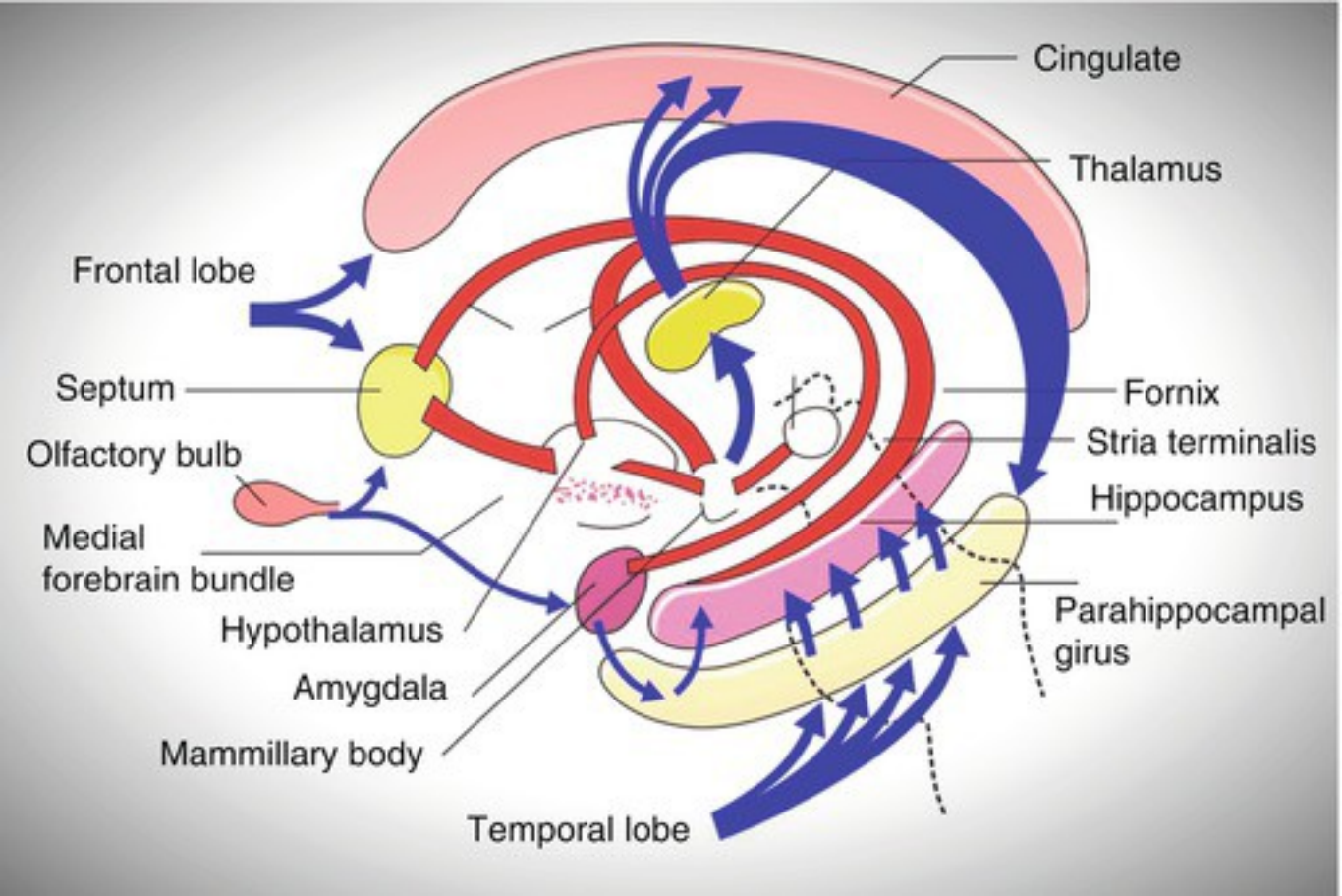


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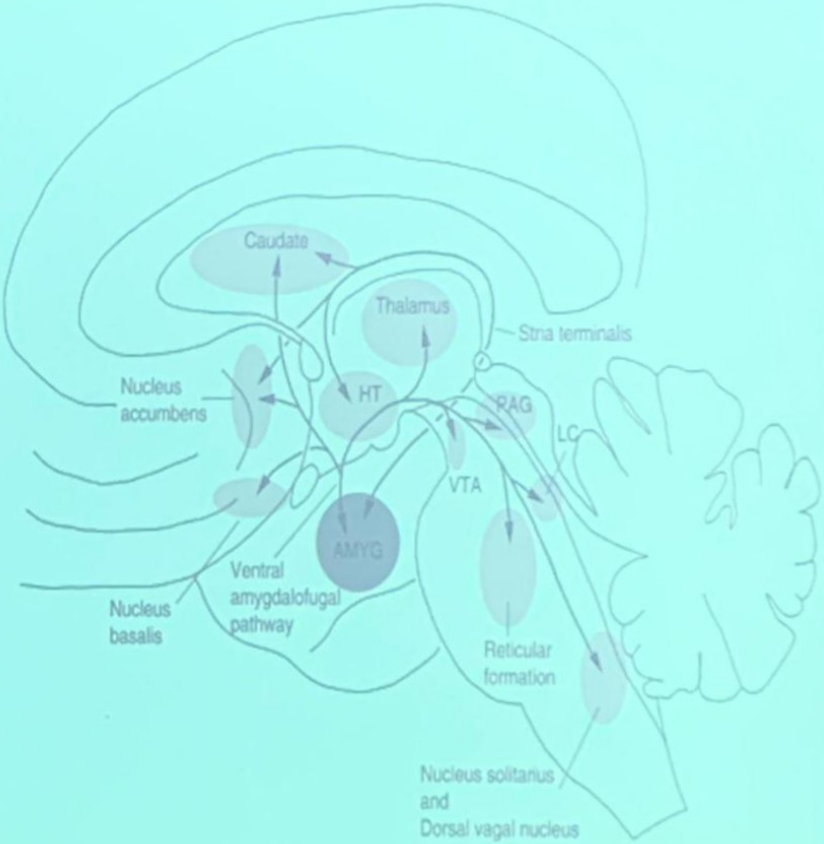
How these pathway integrate: These pathways enable the amygdala to influence hypothalamic functions, such as regulating ANS and neuroendocrine responses. Hypothalamus integrates emotional information received from the amygdala and brain regions then regulates body internal environment (homeostasis) based on emotional state ie. Emotional stress as processed by the amygdala can trigger hormonal and autonomic responses via hypothalamus



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Cingulate
Thalamus
Frontal lobe
Septum
Fornix
Olfactory bulb
Stria terminalis
Hippocampus
Medial
forebrain bundle
Parahippocampal
girus
Hypothalamus
Amygdala
Mammillary body
Temporal lobe

Biological clock and circadian rhythms: Body function vary over the 24hrs of the day i.e. fluctuations of body temperature, sleep and wakefulness are influenced by environmental lightening and mediated by suprachiasmatic nuclei.



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Symptoms of hypothalamus disorders:

Symptoms of hypothalamus disorders vary depending on what hormones are in short supply.

Children might show signs of abnormal growth and abnormal puberty. Adults might show symptoms linked to the various hormones their bodies cannot produce.

There is usually a traceable link between the absent hormones and the symptoms they produce in the body. Tumour symptoms might include blurred vision, loss of vision, and headaches.

Low adrenal function might produce symptoms such as weakness and dizziness.

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Symptoms caused by an overactive thyroid gland include:

sensitivity to heat

anxiety

feeling irritable

mood swings

tiredness and difficulty sleeping

lack of sex drive

diarrhea

constant thirst

itchiness

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